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Invited Review

Repairable inventory theory: Models and applications

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Abstract

Repairable inventory theory involves designing inventory systems for items which are repaired and returned to use rather than discarded. Such systems are composed of items which are typically less expensive to repair than to replace, and are considerably more complicated than traditional consumables inventory systems. The typical problem is concerned with the optimal stocking of the repairable parts and the location of these stocks, given that there may be multiple locations. An added dimension to the problem is the determination of the size and location(s) of the repair capacity for these parts. Further, different performance measures may be used, such as cost, backorders, and availability. There are many complicating factors in the design of repairable inventory systems, for example, not all failed units can be repaired and put back into service, some will be condemned and have to be replaced by new procurements. Various solution approaches have been developed to solve the problem, few have been implemented in practice, and no single model has addressed all or most of the complicating factors. Recent trends in the repairable inventory environment, environmental trends and regulations, and trends in product design are calling some of the assumptions of earlier models into question. In this paper we discuss the existing body of literature on repairable inventory, examine the various models proposed and the major assumptions made in those models, and classify them according to their solution methodology, single versus multi-echelon, and exact versus approximate solutions. It is intended to aid practitioners and researchers in identifying the sources for existing methods and the suitability of those to their application, as well as identify areas for additional research. © 1997 Elsevier Science B.V.

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1. Overview

Repairable inventory theory, unlike classic inventory theory, is concerned with items which are repaired (returned to useable condition) rather than discarded. Repairable inventory systems are typically composed of high cost, long-life goods, where is it less expensive to repair an item than it is to replace it. These repairable inventory systems are common in the military; e.g., aircraft and aircraft components, and in a variety of commercial settings; e.g., copying machines, transportation equipment, and electronics. This type of inventory system is considerably more complicated than a traditional consumables inventory system since there are a number of decisions to consider even for a single-echelon system. Some complicating factors include: demand for units may

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Table 1
Solution methodology

Exact	Approximate	Simulation	DSS
Allen and D'Esopo, 1968	Albright, 1989	Buyukkurt and Parlar, 1993	Malmborg et al., 1987
Albright and Soni, 1988	Albright and Soni, 1988	Dhakar et al., 1994	Petrovic and Petrovic, 1992
Balana et al., 1983	Cheung and Hausman, 1993, 1995	Fisher and Brennan, 1986	
Ebeling, 1991	Graves, 1988	Gross et al., 1984	
Gross and Miller, 1984	Graves, 1985	Hamann and Proth, 1993	
Gross et al., 1983	Gross et al., 1993	Matta, 1985	
Haber and Sitgreaves, 1975	Gross and Ince, 1978	Hausman and Scudder, 1982	
Jung, 1993	Gross et al., 1977	Miller, 1974	
Lee and Moinzadeh, 1987	Gross et al., 1987a,b	Pyke, 1990	
Mabini et al., 1992	Gupta, 1993	Scudder and Hausman, 1982	
Miller and Modarres-Yazdi, 1978	Gupta and Albright, 1992	Scudder, 1984	
Muckstadt and Isaac, 1981	Haber et al., 1975		
Nahmias and Rivera, 1976	Howard, 1984		
Lawrence and Schaefer, 1985	Kim et al., 1996		
Porteus and Lansdowne, 1974	Lee, 1987		
Richards, 1976	Lee and Moinzadeh, 1989		
Schaefer, 1989	Moinzadeh and Lee, 1986		
Schrady, 1967	Muckstadt, 1978, 1979		
Shanker, 1981	Schaefer, 1983		
Sherbrooke, 1968	Sherbrooke, 1971, 1986		
Simon and D'Esopo, 1971	Silver, 1972		
Simon, 1971	Smith and Schaefer, 1985		
Simpson, 1971, 1978			

not be balanced with returns of repairable units; the choice of procurement policies of replacement units, any policy must take into account condemnation or recoverability rates; the choice of review (periodic or continuous) of inventory status; the choice of repair returns (batch or one-for-one), the choice of repair policies (batch or one-for-one).

The repairable inventory problem is typically concerned with the optimal stocking of parts at bases (or forward locations) and a central depot facility which repairs failed units returned from the bases while providing some predetermined level of service. The classic repairable problem is the military logistics problem of stocking repairable parts for aircraft at bases which are capable of repairing some, but not all broken parts, and a central depot which serves all of the bases. Components on aircraft fail and lack of any component or part will render an aircraft unavailable for a mission. The objective of such a system is typically to maximize the availability of aircraft, or conversely minimize shortages and hence the number of grounded aircraft, subject to a budget constraint. While the repairable inventory problem has its roots in military applications (Sherbrooke, 1968) there are numerous possible commercial applications such as the leasing of office equipment (e.g., reproducing equipment and computers), heavy equipment (e.g., tractors, earth moving equipment, industrial presses), and transportation equipment such as railroad, subway cars and buses. The repairable inventory problem is extremely relevant today for both the military and commercial sectors. The challenge for organizations today is to do more with less and repairable inventory theory seeks to maximize the service level while providing the most effective stocking quantities.

Repairable inventory is not the same as spares inventory, although both may serve the same purpose – to provide for the availability of equipment in the presence of failures. Spare parts are not necessarily repairable and spares setting problems often do not consider return flows of units or the repair of units. In this paper, we do not include any spares models, unless the model was designed for spares that are repairable items.

We begin our discussion of repairable inventory theory with a taxonomy for the models and then discuss the simplest class of models, single-echelon, first. In following sections multi-echelon models are discussed, followed by application and implementation works. The intent of this work is provide for an understanding of the theory and its application and to hopefully lead to new insights for future developments in repairable inventory theory.

Table 2
Single echelon models

Author	Model – deterministic or stochastic	Inventory policy
Allen and D'Esopo, 1968	Stochastic	Modified EOQ
Balana et al., 1983	Stochastic	$(s-1,s)$
Dhakar et al., 1994	Stochastic	$(s-1,s)$
Ebeling, 1991	Stochastic	$(s-1,s)$
Fisher and Brennan, 1986	Stochastic	$(s-1,s)$
Graves, 1988	Stochastic	$(s-1,s)$
Gross and Ince, 1978	Stochastic	$(s-1,s)$
Gross et al., 1977	Stochastic	$(s-1,s)$
Haber et al., 1975	Stochastic	N/A
Hamann and Proth, 1993	Stochastic	Service level based
Lawrence and Schaefer, 1985	Stochastic	$(s-1,s)$
Mabini et al., 1992	Deterministic	Modified EOQ
Nahmias and Rivera, 1976	Deterministic	Modified EOQ
Schaefer, 1983	Stochastic	$(s-1,s)$
Schaefer, 1989	Stochastic	(s,s)
Schrady, 1967	Deterministic	Modified EOQ for procurement
Smith and Schaefer, 1985	Stochastic	$(s-1,s)$
Muckstadt and Isaac, 1981	Stochastic	(Q,r)

Table 3
Multi-echelon models

Author	Inventory Policy	Features
Albright, 1989	$(s-1,s)$	Deterministic travel times, machine repair queueing model
Albright and Soni, 1988	$(s-1,s)$	Continuous-time Markov process
Buyukkurt and Parlar, 1993	$(s-1,s)$	Finite machine, repair facilities
Demmy and Presutti, 1981	$(s-1,s)$	METRIC, MOD-METRIC, limited repair funds, variable safety levels
Cheung and Hausman, 1993	$(s-1,s)$	Multiple failures, cannibalization allowed
Graves, 1985	$(s-1,s)$	Negative binomial distribution
Gross et al., 1993	$(s-1,s)$	Zero travel times, finite population, finite servers
Gross et al., 1987a	$(s-1,s)$	Finite source, finite repair capacity
Gross et al., 1987b	$(s-1,s)$	System availability measure
Gross and Miller, 1984	$(s-1,s)$	Two-stage procedure, comparison to DYNA-METRIC
Gross et al., 1984	$(s-1,s)$	Monte Carlo simulation
Gross et al., 1983	$(s-1,s)$	Closed queueing network
Gupta, 1993	$(s-1,s)$	Multi-indentured model, single base
Gupta and Albright, 1992	$(s-1,s)$	Multi-indentured model, multiple bases
Haber and Sitgreaves, 1975	$(s-1,s)$ base, (s,S) intermediate level	Intermediate echelon stockage
Howard, 1984	(s,S)	Batch repair policy
Jung, 1993	$(s-1,s)$	Time varying demand, increasing reliability
Kim et al., 1996	$(s-1,s)$	Quadratic shortage cost function
Lee, 1987	$(s-1,s)$	Transshipments allowed
Lee and Moinzadeh, 1989	Base $(s-Q,s)$, Depot $(s-Q,nQ)$	Batch model
Lee and Moinzadeh, 1987	Base $(s-Q,s)$, Depot (s − Q, nQ)	Condemnation allowed
Matta, 1985	Both $(s-1,s)$ and (Q,r) examined	Simulation model
Miller, 1974	$(s-1,s)$	Distribution problem only – Look Ahead policy developed
Miller and Modarres-Yazdi, 1978	$(s-1,s)$	Distribution problem only – optimality of Look Ahead policy when the number of bases is large
Moinzadeh and Lee, 1986	Base (s_i,Q), Depot (s_0,Q)	Use Batch when set-up cost is high compared to holding costs; One-for-one policy when repair times are long
Muckstadt, 1978	$(s-1,s)$	METRIC approximations, high demand items
Muckstadt, 1979	$(s-1,s)$	Three-echelon model, high demand items
Muckstadt and Isaac, 1981	$(s-1,s)$	Returns not correlated to failures
Porteus and Lansdowne, 1974	$(s-1,s)$	Management controllable repair times
Pyke, 1990	$(s-1,s)$	Simulation, 3 echelon, transshipments allowed, cannibalization allowed
Richards, 1976	(nQ,r) – Procurement only	Optimal replenishment quantities
Hausman and Scudder, 1982	$(s-1,s)$	Simulation, priority repair scheduling
Scudder and Hausman, 1982	$(s-1,s)$	Simulation, spares stocking w/dependent es repair times
Scudder, 1984	$(s-1,s)$	Simulation, priority repair scheduling – multiple failure case
Shanker, 1981	Base $(s-1,s)$, Depot (s,S)	Deterministic repair times
Sherbrooke, 1968	$(s-1,s)$	METRIC formulation
Sherbrooke, 1971	$(s-1,s)$	Min E[aircraft not available], cannibalization allowed
Sherbrooke, 1986	$(s-1,s)$	Multi-indenture model, pipeline variance
Silver, 1972	$(s-1,s)$	Max E[ready rate of aircraft], cannibalization allowed
Simon, 1971	Base $(s-1,s)$, Depot (s,S)	Steady state inventory equations
Simon and D'Esopo, 1971	Base $(s-1,s)$, Depot (nQ,r)	Exact distribution of steady state inventory
Simpson, 1971	(Q,r) Base procurement	Optimal replenishment quantities

Table 4
Applications models/implementation

Author	Problem	Solution technique	Features
Ashayeri et al., 1996	Management of repairable spares for PCs	Analytic and simulation	Provides for structured management decision making and policy setting
Davidson, 1980	Commercial airline spares setting	Case study	Evaluation of three spares models
Demmy and Presutti, 1981	USAF initial provisioning and replenishment	MOD-METRIC, METRIC	Budget constraints, variable safety levels
Howard, 1984	Mining equipment	s,S type model	Uses standard inventory theory, units batched
Kharola and Vrat, 1986	Bus fleet spares	METRIC-type	Two-echelon, scheduled maintenance and failures
Kumar and Blumberg, 1991	Mining equipment	Finite population queuing	Estimates E[Total annual costs]
Kumar et al., 1994	Diesel locomotive spares	DSS using physical system theory and systems dynamics	Evaluates effects of production, repair, storage and transportation policies
Singh et al., 1980	Bus fleet spares	METRIC-type	Failure rates based on geographic area of route and average mileage
Tedone, 1989	Commercial aircraft availability	DSS	Incorporates shortage costs which consider the cost of a canceled flight

This review is limited to articles published in academic journals, and while we are aware of other work which has been done, it has been omitted since it may not be readily available world-wide. This review is intended to be as comprehensive as possible, but we recognize that it is almost impossible to be truly comprehensive and accept this limitation. We acknowledge and recognize the contribution of earlier reviews of repairable inventory theory by Nahmias (1981) and Mabini and Gelders (1991). Nahmias (1981) provides a comprehensive review, but includes spares models in his review. Mabini and Gelders (1991) provide only a partial review of the research in print.

2. A framework for understanding

We have attempted to provide a logical framework for understanding repairable inventory theory by classifying the work by a number of dimensions. Table 1 describes the predominate type of solution methodology employed by the author(s), Optimal, Approximate, Simulation or Decision Support System (DSS). Table 2 partitions the single-echelon models into deterministic or stochastic type models and then further differentiates the work based on the type of inventory policy used. Table 3 provides a similar type of classification for multi-echelon models, but since models are stochastic, an added dimension, the main feature of the model, is now used. Features, in this case, refers to the major contribution of the model; e.g., allowing cannibalization of parts. We will discuss single-echelon and multi-echelon in separate sections, but the discussion of solution methodologies will be interspersed throughout both sections. In Table 4 applications and implementation of repairable inventory theory models are classified by the type of problem addressed, the solution technique used and features of the models. Since there are relatively few implementations reported in the literature, a separate section is devoted to this important area of repairable inventory theory.

3. Single-echelon models

Repairable inventory theory, as stated earlier, is more complex than consumable inventory theory and numerous authors have chosen to focus their efforts on the single-echelon problem. Since focusing on the single-echelon problem removes some of the com-

plexity, most of the work in single-echelon theory is concerned with optimization of a particular criterion. Typically the single-echelon problem focuses on optimal replacement quantities for repairable parts that are worn beyond recovery or have been lost from the system or on optimal batch sizes for repair lots. There are two basic formulations of the single-echelon problem: (1) deterministic, batch-procurement (EOQ-type) models, or (2) stochastic, one-for-one replenishment models or batch models.

3.1. Deterministic single-echelon models

Schrady (1967), Nahmias and Rivera (1976) and Mabini et al. (1992) all develop models which assume deterministic demand and known, constant recovery rates. Schrady's work focuses on optimal procurement quantities by using a modified EOQ formulation, given a constant repair batch size where the goal of the system is to minimize the cost of operating the repairable inventory system (Schrady, 1967). No backorders are allowed and the repair facilities are assumed to be uncapacitated. Nahmias and Rivera (1976) extend Schrady's model for finite repair rates and determine the optimal repair batch size in conjunction with optimal procurement quantities. Mabini et al. (1992) also derive a modified EOQ model and examine the interactions between procurement and repair, but their formulation allows backorders.

All of the single-echelon deterministic models consider the case of a single item considered only on the end item level. Many products which make up a repairable inventory system are complex items with a multi-indenture structure. Holding additional units of an multi-indenture item at only the end item level is most likely holding inventory at the most expensive level, from both a holding cost and procurement cost perspective. Recent design trends have also lead to modular designs for many repairable items, such as commercial aircraft engines, and any optimal inventory model should expressly take into account the impact of these changes on inventory stocking decisions. Another limiting assumption of this class of models is a known, constant condemnation rate. Many repairable inventory systems are composed of items of various ages being subjected to differing conditions (i.e., geographic location of use and annual time in service) which would lead to variable failure rates and condemnations. As with all models which require estimates of holding costs, procurement costs, and shortage costs, the estimates used may not hold for all types of repairable items and the sensitivity of the models to changes in the cost parameters is not evaluated in any of the studies.

3.2. Stochastic, single-echelon models

Stochastic single-echelon models deal with situations where demand rates vary over time. Allen and D'Esopo (1968) focus on order policy to replace non-repairable failed items. They use a reorder point, order quantity replenishment policy. The analysis is limited to a single item, and the model allows for varying backorders, based on return to assets of repairables from repair. Muckstadt and Isaac (1981) deal with the interesting problem of return rates which do not correspond to failure rates (demand rates). The system developed seeks to minimize expected backorders in the system with a optimal (Q,r) procurement model and an uncapacitated FCFS repair shop. Dhakar et al. (1994) examine the case of high cost, low demand critical repairable items. Their system allows for no condemnations or cannibalization. The one for one inventory replacement policy $(S-1,S)$ is used in conjunction with three modes of replenishment (lead times): normal, emergency, and expediting. A simulation approach is used in their analysis.

In another batch policy paper, Schaefer (1989) derives an optimal (s,S) procurement policy to minimize the expected ordering, holding and shortage costs. All procurement lead times are assumed to be either zero or deterministic. Simpson (1971) derives a batch ordering model (Q,r) for new stock for a repairable inventory system assuming that the mean demand rate exceeds the mean repair rate. The model is based on the availability of the repair facility to meet a schedule and Simpson (1971) observes that if the variance of the output of the repair shop is decreased then the average inventory may be reduced while providing the same level of service. The model is limited to a single item with a single level of indenture structure and that unfilled demand is backlogged.

Schaefer (1983) presents a dynamic programming formulation to maximize the job completion rate, subject to a holding cost constraint, for a single item composed of a single part with independent repair times. No backorders or condemnation of units are allowed. Lawrence and Schaefer (1985) propose another dynamic programming formulation for calculating optimal inventory levels to handle the multiple item case where the goal is to minimize the expected holding costs and shortage costs. No backorders are actually allowed, but rather the system is assessed a penalty for a shortage. Smith and Schaefer (1985) present a heuristic for the Lawrence and Schaefer (1985) problem and modify the heuristic for the case of increasing failures rates over time.

Fisher and Brennan (1986) examine various cannibalization policies to answer the question whether cannibalization should be done and when, using a simulation approach. A single echelon, single station model is used, with a stated performance measure of minimizing backorders. Further, an infinite calling population is assumed, along with ample servers. The findings are that the best cannibalization policies are those that allow cannibalization only when it can be done quickly relative to repair, or without delaying part repair actions. The policy of complete cannibalization is found to generally perform poorly. The addition of resource constraints (e.g. limited servers) further alters the desirability of complete cannibalization. Graves (1988) examines the case of multiple failures in determining resource requirements and inventory levels. A discrete-time, linear-systems model is developed, and is implemented with a spreadsheet package. Assumptions made include small known lead times, deterministic repair times, independent failures, ample component spares, and large calling population. Haber et al. (1975) develop demand prediction techniques for repairables. The forecasting is at a single level, and an analytic model is developed for low demand items, subject to sporadic demand. The model was tested on data obtained from Navy patrols. The proposed procedure provides better stocking policies than conventional methods of estimating line item demands. Hamann and Proth (1993) use a simulation approach to model the repairable tools case. Using the global information available, they do a K-means analysis to reduce the problem size and create homogeneous classes. They derive behavior of the tool types from the results of the study and fit probability distributions to the operating and repair periods.

There are some models which consider finite source and finite repair capacity. This then translates into solving the classic machine repair problem of queuing theory, or some variant of this problem. Balana et al. (1983) determine the optimal levels of spares and repair channels. They consider the case of finite source queue and finite repair channels and the system is in a transient environment. The solution approach is to solve for the classic machine repair problem of queueing theory, which is the simplest case of a repairable inventory system. Two performance measures are considered: Generalized Availability, and Expected Backorders. Ebeling (1991) uses a queueing approach (M/M/s finite population queue) to determine the optimal allocation of stock levels and repair channels. Subsequently, dynamic programming is used to allocate a fixed budget among the multiple items in order to maximize system availability considering the optimum trade-off between number of repair channels and item inventory levels. Gross and Ince (1978) determine the number of spares and repair channels for items subject to stochastic failure. They extend the classic machine repair problem with spares to include more than one step in the repair phase, and treat it as a cyclic queue. An exact model is compared with an approximation. They find that true availability of the system is greater than that predicted by the approximate model. For high availability of spares (light traffic queues) the approximate model is found to be accurate and computationally efficient. Again, the analysis is limited to a single item, closed system (no condemnations), and infinite queue lengths. Gross et al. (1977) determine the number of spares and repair lines for replacing and repairing components which randomly fail. They set their model for a 90% fill rate and minimize system cost subject to that fill rate. The queueing model determines the fill rate for a given level of spares and then is imbedded in a heuristics cost optimization model. They assume steady state, even when population size and component reliability change, an instantaneous change to steady state is assumed rather than transient state. The implication is that for most provisioning purposes, failures are frequent. Further, the heuristic

cost minimization model operates on a year by year basis, and has the limitation that it does not look ahead.

The stochastic models suffer from many of the same limitations as the deterministic models. The models all are limited to a end item (or a set of end items) which are composed of a single unit, which ignores the problem of multi-indentured items. Most models also assume independent repair times, which is also referred to as the sufficient channels assumption, which simply assumes that repair facilities are uncapacitated. This may or may not be a realistic assumption in a time when military repair depots, as well as their commercial counterparts, are being downsized and planned utilization rates of 95% are not uncommon. An additional limitation of the models developed by Schaefer (1983, 1989), Lawrence and Schaefer (1985), and Smith and Schaefer (1985) is that only 'ultra-reliable' avionics units are considered. These units are fairly simple from a manufacturing repair perspective, since often repair operations are limited to a single machine. The units are also assumed to experience a very small number of field failures and most of the repair operations may be viewed as a form of periodic maintenance. In this case, the items may essentially be viewed as new when they finish the repair process, since repair here actually involves replacement of parts which have not actually failed. As with the deterministic models estimates of various cost parameters are required. Many of the stochastic models assess a backorder cost, which may be very difficult to determine. For example, the cost of a grounded commercial aircraft must consider the lost revenues from the flight non-availability, including all lost connections downstream, the cost of a crew cancellation, the positive saving from no fuel expenditures and the intangible costs in terms of lost goodwill. These penalty costs become even more difficult to assess when military assets are involved and the question of the cost of non-availability changes from peacetime to war. Finally, the use of dynamic programming as an optimal solution technique where there is only a single item or a very limited set of items to consider is computationally reasonable, but in the case where there may be thousands of items the computational time may well be unrealistic, even by today's computing standards.

4. Multi-echelon models

The typical multi-echelon problem is a two-level system composed of a number of bases being served by a central depot, or depots. Failures of operating units at the base-level generates demand for units, which may be satisfied by base-level stock, and at the same time the failed unit will be placed into the repair at the base-level repair facility or at the depot-level repair facility. In the event sufficient stock is not available at the base-level, a backorder occurs and demand must be satisfied from units at the depot-level, either from existing stock or from units completing the repair process. The system, if no condemnation of units occurs, is a closed system. The problem, in the simplest form, is the setting of repairable inventory levels at each of j bases such that a prespecified level of service is achieved, given repair rates and transportation times and a budget constraint. If condemnations are allowed, then the model must plan for procurement quantities and timing. Additional complications may arise such as allowing transfers of units from one base to another to satisfy demand (the transshipment problem), cannibalization, re-distribution of repaired units among the bases, capacity constrained repair facilities, batching of repair and replenishment quantities, and various other factors.

4.1. METRIC

Many repairable inventory theory models are based on Sherbrooke's METRIC model for setting repairable inventory levels and allocating these units so as to achieve some desired level of service, measured by expected backorders at the base level (Sherbrooke, 1968). Later models such as MOD-METRIC and VARI-METRIC are extensions to the base METRIC model and many modifications, such as allowing batch repairs or lateral shipments are based on METRIC logic. METRIC takes a system view of the repairable inventory problem since it is concerned with setting the initial levels of inventory and the distribution of the inventory among j bases so as to support a system-wide objective of minimizing backorders. We provide a complete description of METRIC since it represents a fundamental development in repairable inventory theory and because

so many later models are METRIC-based. A full explanation of the METRIC logic follows.

METRIC is based on the following assumptions:

1. The demand for units at each base is a stationary Poisson process.
2. There is no lateral resupply (transshipments) among bases.
3. There are no condemnations of items (all items are repairable).
4. All items have the same essentially.
5. No delay or batching of items at the repair facility before items start repair.
6. Repair times are statistically independent.
7. The probability of a failure of one part is independent of failures occurring for other parts.
8. Each failure repaired is caused by a failure of at most a single part/component.

In addition, METRIC requires that the following information be available:

1. the variance-to-mean ratio of demand;
2. optimization targets, expressed as the maximum dollars available and/or the maximum expected backorders per item allowed;
3. the average depot repair time for each i item;
4. the item unit cost for each i item;
5. the average fraction of each i items which are recoverable;
6. the average jth base repair time for each i item;
7. the average order and shipping time from the jth base for the ith item;
8. the demand observed over the time period n;
9. forecasts or estimates of planned flying hours.

METRIC can be stated as follows: let $\beta_{ij}(s_{i0}, s_{ij})$ be the expected backorders for item i at base j, depot stock level is s_{i0} and the stock at base j is s_{ij} and c_i is the unit item cost, then the problem is to minimize the cost of repairable spares subject to upper limits on the number of allowable stockouts, or:

$$\text{Minimize} \quad \{s_{ij}\} \sum_{i=1}^{I} c_i \sum_{j-0}^{J} s_{ij}$$

$$\text{subject to} \quad \sum_{i=1}^{I} \beta_{ij}(s_{i0}, s_{ij}) \le b_j.$$

The METRIC model logic may be explained most easily in five steps (Sherbrooke, 1968). In the first stage the average time required for a base request for resupply from the depot is computed, this delay will need to be calculated for each of j bases. Consider the case where there are always sufficient units of stock at the depot to satisfy demand ($s_0 \to \infty$), there will be no delay between the time a resupply order is received and shipped. The average delay time will be only the order and shipping time, O_j. If, on the other hand, the average stock at the depot is some quantity δ, then there will be some delay due to the average repair time D and the resupply time will be $O_j + D$. The delay at the depot can be expressed as a function of s_0; if there are s_0 or less units in depot repair there will be no resupply delay but if there are greater than s_0 in repair, then the resupply of $x - s_0$ units will be delayed. The expected number of units on which delay will be incurred may be expressed:

$$B(s_0 \mid \lambda D) = \sum_{x=s_0+1}^{\infty} (x - s_0)\, p(x \mid \lambda D).$$

This is the expected number of units backordered at the depot. This is not an explicit part of the objective function which seeks only to minimize the sum of backorders at the bases (which directly result in a non-available asset). The total expected delay over any time period may be calculated as the expected number of units on which delay will be incurred, $B(s_0 \mid \lambda D)$, multiplied by the length of the time period of interest.

In the second step, for each level of depot stock, s_0, and each base the expected backorders must be computed as a function of the base stock, s_j. The expected backorders as a function of the base stock is given by:

$$B(s_j) = \sum_{x=s_0+1}^{x=\infty} (x - s_j)\, p(x \mid \lambda_j T),$$

$$T = r_j A_j + (1 - r_j)\left[O_j + \delta(s_0) D\right],$$

where $\delta(s_0) = B(s_0 \mid \lambda D)/B(O \mid \lambda D)$, A_j = average base repair time, and r_j = average fraction of units that are base repairable.

In step three for each level of depot stock, s_0, the optimal allocation of the units of stock to each of the J bases must be determined so as to minimize the

sum of expected backorders at all bases. This can be easily done via marginal analysis by adding the next unit of stock to the base where it results in the greatest decrease in expected backorders, $B(s_j)$.

In step four, for each level of constant total system stock, $s_0 + s$, select the minimum expected system backorders. In step five, the multi-echelon problem is addressed by marginal analysis. The next investment is allocated to the item which produces the maximum decrease the expected backorders divided by unit cost.

METRIC is based on the assumption that the length of backorders and the number of backorders are linearly related, and the primary reason for this assumption is tractability (Sherbrooke, 1968). Another important consideration for the use of METRIC relates to the linear structure of costs, which assumes that all cost structures, such as backorder costs and holding costs, follow strictly linear relationships. An important assumption inherent in METRIC and METRIC-based formulations is that of perfectly correlated failure and demand for units. In this case the failure of a unit immediately generates the demand for a repaired unit, with no time lag, and in an $(s-1,S)$ system this is a reasonable assumption. However, when a batch system is being used this assumption may not be reasonable. In a later section, we discuss a model developed to cope with this problem.

METRIC does not allow for the condemnation of units and METRIC is also limited to a units consisting of a single part, (a single level of indenture) and as a result may suffer from the limitations as single echelon models with respect to costs. In part, VARI-METRIC was formulated by Sherbrooke (1986) to deal with the problems associated with multi-indenture units, and these modifications to METRIC will be discussed in a later section.

4.2. Metric-based extensions and modifications

Sherbrooke (1971) provides a reformulation of METRIC, to allow for the objective function to directly minimize the expected number of non-operationally ready aircraft. However, this change in the objective function no longer ensures optimality and makes the solutions provided only approximate. Muckstadt (1973) develops an extension to METRIC, known as MOD-METRIC which allows for multiple levels of indenture, i.e., spares requirements for an end item and its components. However, MOD-METRIC becomes increasingly complex as the number of levels of indenture grows; Muckstadt (1973) limits discussion to a two echelon example. Muckstadt (1978) proposes a Lagrangian optimization technique to solve the MOD-METRIC formulation at a much faster computational speed. The approximation method makes using MOD-METRIC for very large problems more practical, while providing a very close to optimal solution. Muckstadt's comparisons of the approximation technique is limited to high demand items and it is unclear whether or not very expensive units, such as aircraft engines, would be subject to high demand on the end-item level. Muckstadt (1979) also extends METRIC to handle the case of a three-echelon supply system, where there is an intermediate maintenance location in addition to the bases and a central depot, and all products are assumed to be a modular design. The work does provide for some examples of determining optimal stocking levels using assembly families, but the assemblies are limited to a single part failure. This three-echelon model is based on the behavior of high demand avionics components subject to a single failure. Again it is not clear how low demand items, multiple failures or multiple levels of indenture would effect the final allocation of stocks.

Simon (1971) derives the exact expressions for inventory positions and allows for condemnations and examines METRIC's approximations as a special case of his work. However, the model of Simon (1971) does not relax any of the assumptions of METRIC and suffers from the same limitations.

Shanker (1981) provides exact expression for a system which is less general than METRIC since it assumes deterministic repair times. Shanker (1981) examines batch and unit inspection policies, both of which allow condemnation of units. All of the assumptions of METRIC hold in this analysis, in addition to the new ones added.

Lee (1987) provides a METRIC-based approximation for a two-echelon system where some lateral transshipments are allowed. Lateral transshipments are where items backordered at a base may be transferred from stocks at another base, provided sufficient stock exists at another base and the bases

are within the same pooling group. Lee (1987) examines three possible rules for choosing bases for the transshipment of units to originate. The model assumes that all repair operations are done at the depot level only and that a unit is composed of a single unit (single level indenture). The model seeks to minimize total costs, composed of holding costs, backorder costs, and the cost of transportation, rather than backorders. No condemnation of units is allowed and high fill rates (> 0.70) are required at all bases. The requirement that repair is done only at the depot level ignores the possibility of base level repair operations and the resulting shorter repair lead times (including transportation times). The decision criteria for how to actually pool bases for transshipments is not covered and is a potentially important topic when considering such a system. A additional restriction on the model is that all bases in the same pooling group have the same repair arrival rates and the effect of bases with different repair arrival rates is not discussed. There is also no analysis of the sensitivity of the model to changes in the various costs or the impact of lower fill rates at the base level. The model is also limited to a single indenture product, as are the majority of METRIC-based formulations.

Demmy and Presutti (1981) review applications of multi-echelon inventory theory in the Air Force Logistics Command. They discuss METRIC step by step, discuss MOD-METRIC, LMI Procurement model and consider a procurement-repair model (limited repair funds). In this scenario, nonservicable assets may be held in a pool to be repaired later if funds are not available at the current time. MOD-METRIC is discussed as a method for initial provisioning of spares, while for replenishment spares, METRIC based Variable Safety Level is utilized. Silver (1972) presents a model developed for the Canadian Air Force which uses a $(s-1,s)$ inventory policy based on METRIC, but with the objective of maximizing the ready rate of aircraft (ready rate here is defined as the probability of zero backorders). The formulation of Silver (1972) allows cannibalization, but uses all of the same restrictive assumptions as METRIC. The model is limited to products with single level of indenture and because of its maximization structure only provides approximate solutions.

Because of the shortcomings of METRIC, two models we discuss here were formulated: DYNA-METRIC and VARI-METRIC. VARI-METRIC addresses the problems of multiple indenture products and the variance of pipeline inventories. DYNA-METRIC deals with the problem of surges in the demand (i.e., non-steady state conditions) for repairable parts during wartime.

VARI-METRIC was developed by Sherbrooke (1986) to provide more accurate estimates of the true number of backorders experienced by bases in a multi-indenture, multi-echelon repairable inventory system and to improve on the computational speed of METRIC. Like METRIC, the model uses a continuous review $(s-1,s)$ system and prohibits condemnation of units and any form of transshipment. It does provide an approximation, rather than optimality, due to the nature of the problem. In the case of multi-indenture products the end item is restricted to, at most, the failure of a single component, composed of a single part. Repair operations are deterministic in duration and may not be delayed for a lack of spares. VARI-METRIC provides much more accurate estimates of expected backorders than the METRIC or MOD-METRIC models which tend to understate the expected backorders.

DYNA-METRIC is based on the theory that during wartime military repairable inventory systems will be subject to demand surges (non-steady state behavior), so that variance to mean ratios may be greater than one. The model does assume however that failure rates are stationary. DYNA-METRIC is a simulation model concerned with spares stocking, priority repair and distribution and the model seeks to make decisions which minimize the expected number of grounded aircraft. It does allow transshipments, cannibalization and does not assume that ample server channels will be available. Pyke (1990) evaluates DYNA-METRIC for a three-echelon system, using only electronic components, and comes to a number of important conclusions. Pyke (1990) observes that a very small increase in repair times leads to a very large increase in the number of expected grounded aircraft and this leads to two general observations: first, repair shop utilization rates should be no greater than 0.80; and two, if transshipments are used, shorter transportation times provide the maximum benefit. Pyke's evaluation was

limited to single indenture products composed of a single part (Pyke, 1990), as DYNA-METRIC itself is restricted too. Since DYNA-METRIC is a simulation-based model, only approximate solutions are available.

5. Multi-echelon batch models

All of the multi-echelon models discussed to this point have been based on one-for-one repair and replenishments. There are, however, cases where it has been determined that batching repair and/or procurement of units may be more economical. All of the work presented in this section is based on the premise that METRIC's assumptions hold and all of the models presented are for two-echelon systems.

Moinzadeh and Lee (1986) show that, under certain conditions, it is less expensive to use a batch replenishment policy. These conditions are satisfied when: (1) the set-up costs of ordering and shipment are high relative to the holding cost of the item, (2) the repairable items are subject to high demand, (3) there are few bases in the multi-echelon system, (4) average repair times are lengthy, and (5) the transportation time from the depot to the bases is long. Additionally, Moinzadeh and Lee (1986) require that no repair is possible at the base level, as well as the common assumptions such as sufficient repair channels, no condemnation of units, and no transshipments allowed. The supposition made with respect to the ratio of set-up to holding costs is interesting in that repairable items have long been considered to be very expensive assets and that holding costs are normally a function of the value of an item. There are no examples given of an actual case of such a product.

Lee and Moinzadeh (1987) show that a one-for-one $(s-1,s)$ replenishment policy is a special case of the general batch system $(s-Q,s)$ where the optimal batch size, Q, is equal to one. Condemnations are allowed in this system, but no repair operations are done at the base level and sufficient repair channels exist at the depot. Approximate solutions are derived for stocking levels by solving the model in order to minimize the expected backorders. This approximation does require the estimation of holding costs and backorder costs, the problems of which have been discussed in Section 4, in addition, no large problems were solved and the computational effectiveness of the approximation is unknown.

In a final work Lee and Moinzadeh (1989) present optimal stocking and reorder point levels for a batch system with random, independent and identically distributed (IID), distribution and repair times. All inventory is assumed to be held at the repair center (depot) and condemnation of units is allowed.

Richards (1976) develops (nQ,r) replenishment policies for a generalized two-echelon system where condemnations are allowed. The major assumption of this model is that the processes which generate repairable and non-repairable failures are independent and stationary. Richards' major finding is that decreasing the lead times for procurement and repair lowers the probability of stockouts and the expected number of backorder days. The difficulty of estimating the costs, especially stockouts, required for modeling a repairable inventory system is explicitly recognized by the author, but no attempt is made to evaluate the sensitivity of the model to error rates in cost parameters. The model is limited to units consisting of a single level of indenture and the effect of multiple levels of indenture on the model's optimality is not known.

Haber and Sitgreaves (1975) model the optimal purchase quantity for replacements given that not all the failed units turned in will actually be repaired. The model considers a single base, single intermediate resupply, repair echelon, is probabilistic and considers a single item. The expected surplus, expected shortages are computed. The optimal purchase quantity is found to be more sensitive to failure rates than to repair rates.

Simon and D'Esopo (1971) provide for condemnation in an extension to earlier work by Allen and D'Esopo (1968). The exact expression for the stationary distribution of stock on-hand and backordered is derived using a $(s-1,s)$ policy at the base level and a (nQ,r) policy at the depot. and assuming deterministic lead times for all procurement. The model is evaluated for a single item and that item is a single level of indenture.

6. Multi-echelon, distribution-focused

All of the previous works assume that upon completion of repair at the depot a repaired unit is either

placed into inventory or, in the case of backordered demand, sent to satisfy backorders based on a first-come, first-served (FCFS) priority system. This use of a FCFS system for satisfying backordered demand may not be optimal in terms of minimizing the expected number of backorders at all bases served by the depot. Miller (1974) proposes a lookahead policy where the depot allocates the repaired unit to the base which will have the greatest marginal decrease in expected backorders. The objective is to minimize the expected number of backorders for the system and Miller (1974) uses a dynamic programming approximation to test the optimality of his lookahead policy. In a later work Miller and Modarres-Yazdi (1978) show, via marginal analysis, that the lookahead policy provides optimal results when the number of bases approaches infinity. In both cases the results are based on a single unit with a single level of indenture and the effect of multiple levels of indenture is unknown. Both works also assume the ample repair channels exist at the repair facility, as well as the standard METRIC assumptions. Buyukkurt and Parlar (1993) consider a simulation approach to evaluate three allocation policies in a two-echelon model with state dependent failure and repair rates. Two of the three allocation policies are static, while the third is dynamic. Repair capacity is considered limited, and the calling population finite. Several criteria are evaluated in the study. It is noted that interfailure and repair time distributions affect model only in case of high system congestion. A finding of the study is that the dynamic allocation policy is better than the two static allocation policies for three of the five criteria evaluated.

7. Multi-echelon, simulation-based

Given the nature of the repairable inventory problem, it would seem that simulation would be an ideal vehicle for investigating a large number of policies and the restrictive assumptions required by analytic models. Hausman and Scudder (1982) examining the effect of priority scheduling rules in a simulation model of a finite capacity repair shop which serves a repairable inventory network. The simulation model's repair shop is based on an commercial airline's repair facility for jet engines. Initial spares levels are set by using MOD-METRIC and the priority dispatching rules which minimize the expected number of delay days for units at the repair shop are based on inventory needs at the base level. The model examines a product with a three level indenture structure, but uses closed routings and deterministic repair times for each operation. The simulation model also assumes that each unit is limited to single failure and that there are uncapacitated diagnostic and test facilities. The variability of repair lead times, in this case, is limited to the uncertainty in queuing times at the individual work centers at the repair shop. Since the repair facility is subjected to known utilization levels, the variability of lead times in such a system may be very low and as a result not be very different from deterministic repair times.

A related study by Scudder and Hausman (1982) examines the effect of dependent repair times on spares stocking. The simulation model uses Muckstadt's MOD-METRIC to set the initial spares and then uses the finite capacity repair facility described in Hausman and Scudder (1982) to examine the impact of dependent repair time on the actual backorders experienced by the system. The model uses the same three level indenture product structure with each unit limited to a single failure. Repair routings are closed for every part and all repair operation times are deterministic which again limits the variation of repair times to queuing times at the individual machine centers. The study finds that MOD-METRIC sets robust spares levels in the presence of dependent repair times. This finding is interesting since Sherbrooke (1986) finds in his comparison of VARI-METRIC to METRIC that product structures with multiple levels of indenture cause METRIC to underestimate the expected number of backorders in a repairable system even with completely deterministic repair times. A study of the sensitivity of MOD-METRIC to cost parameters would have provided some interesting insights as to the effect of variability on spares robustness. A final simulation study by Scudder (1984) uses the same parameters as the study by Hausman and Scudder (1982), but allows multiple failure of parts. As in the single failure priority dispatching case, local rules which consider base level needs provide higher levels of availability of spares. Since the model has the same parameters

as the earlier works it has the same limitations and drawbacks.

An earlier simulation study by Matta (1985) considers the cases of a limited calling population and increasing or decreasing failure rates. Both $(s-1,s)$ and (Q,r) inventory policies are examined with an uncapacitated repair facility using a FCFS policy for repair demands and condemnations are allowed. Matta's study is the only one which considers the effect of a limited number of items in the system and variable failure rates. The probability of a unit failure may be based on: the number of items in the population, the percentage of new or repaired units in the system or the age of the individual unit. Matta (1985) finds in a sensitivity analysis of the cost parameters that increasing holding costs decreases the optimal inventory quantity, an increase in repair costs increases the optimal inventory quantity, and an increase in procurement costs results in an increase in the optimal inventory quantity. The results are limited to a single item, composed of a single part and no discussion is provided in the paper of the periodic review mechanism used or how spares levels were set. The focus of the work is on the simulation model and its actual insights into repairable inventory theory are limited.

It would appear that, despite its promise, simulation modeling of repairable inventory systems has provided very limited insights. Most models have kept the primary METRIC assumptions or have provided very limited tests of the assumptions. Other authors have used a simulation approach for modeling repairable inventory systems, but we defer discussion of these models until the applications section and note that most of these works have used simulation to verify an analytic model.

8. Multi-echelon, one-for-one models

Porteus and Lansdowne (1974) formulate a $(s-1,s)$ inventory policy for the two-echelon problem with controllable repair times in order to examine the possible trade-offs between spares levels and machining and staffing at the repair center. Their objective function seeks to minimize the expected weighted shortages, subject to a budget constraint. The major conclusion of the study is that in order to reduce shortages either more spares must be added to the system or shorter repair times. Repair times here are modeled as IID random variables and this model is unique in that it explicitly models the repair activities as well as the supply activities (the spares). However, the model is limited to a single indenture product structure, many of the cost parameters used in the model may be highly subjective and all repair activities are limited to a central facility (depot).

Kim et al. (1996) present an approximate solution to optimal spares model designed to satisfy a minimum fill rate at minimum cost (composed of holding and shortage costs). Repair may be done at either the base or depot level, with a high probability (> 0.5 in all cases) of repair being possible at the base level. Unlike previous works, shortage costs are assumed to be of a quadratic form, rather than linear. The total cost function will be unimodal on $[0,\infty]$, since holding costs are linear and shortage costs will be convex. The algorithm developed here has the advantage of computational speed, but no spares are allowed to be held at the depot level and the model assumes an infinite calling population. The model is also limited to a single level of indenture product structure and there is no indication of how sensitive the formulation is to the estimated cost parameters.

Muckstadt and Isaac (1981) present a multi-echelon formulation of the problem of failures not being perfectly correlated to return. The objective is to minimize expected backorders in the system, subject to a budget constraint. A strict $(s-1,s)$ system is used by the bases. The model is limited to a single item, with a single level of indenture structure and assumes uncapacitated repair facilities.

Cheung and Hausman (1993, 1995) examine the two-echelon system with multiple failures incorporated into the model. and present an approximate solution method. The basic assumptions made are similar to the METRIC model. For the multiple failures, partial resupply is not allowed, complete cannibalization occurs at the bases. Upper and lower bounds are developed for the mean number of jobs backlogged at base. Graves (1985) develops a model for a two-echelon system. The assumptions are similar to those of the METRIC model. Further, all repair takes place at the depot only. An exact model for the steady state distribution of inventory at each site is

developed. An approximation is made for the steady state distribution for the case of ample servers at the repair depot.

Most of the models developed along this stream require many of the same assumptions as METRIC. While Muckstadt and Isaac (1981) allowed imperfect correlation between demand and returns, no further work has been done in this potentially valuable area. As repairable systems grow in size (such as many military systems have done) there is often a large time lag between the failure of the unit and a unit being put into a repair facility. The use of supply warehouses are very common and these intermediate locations add a third echelon to the problem. In addition, the commercial use of repairable units is growing and in many cases the consumer may own the units rather than the manufacturer, as in the case of a cellular telephone remanufacturer in the United States. This problem compounds the problem of matching returns with demand and for products with rapidly evolving designs (i.e., cellular phones) where obsolescence may also be an important concern.

9. Multi-echelon queueing based models

In the previous sections, the discussion of multi-echelon models has focused on those where the calling population is assumed to be sufficiently large and there are ample servers (repair channels). These assumptions may be questionable in many situations. Indeed, as military drawdowns occur, such as the ones in the US military, both of these assumptions may not hold true. There is a class of models which has been developed in literature which addresses the multi-echelon repairable inventory system for the case of finite sources and finite repair channels. These models allow for both steady state as well as transient conditions to be analyzed, and are generalizations of the classical machine-repair queueing model. While these queueing models are more realistic than METRIC models, they are also more difficult to solve.

Albright (1989) examines a two-echelon system, comprised of several bases and a central depot, and develops an approximation to the steady state probability distribution of the system for a single item. Standard assumptions are made such as no condemnations, no lateral supply, and deterministic travel times. A First Come First Served (FCFS) priority rule is assumed, and the criterion is to minimize backorders. Albright finds traffic intensities to be a better indicator of whether a system would work well rather than stockage level. A comparison to the METRIC model is also made. The METRIC model is run under the same conditions of finite sources and repair channels and is shown to be rather poor in performance. METRIC almost always underestimates the total backorders in the system. Albright and Soni (1988) model a two-echelon system using continuous-time Markov processes. Since this approach leads to very large state spaces, they utilize aggregate/disaggregate techniques to develop a solution algorithm for finding the steady state distribution. Again, their approach is for a single item, and no spares are kept at the depot.

Gross et al. (1993) provide solution methods for the steady state probability distributions. The multi-echelon inventory system is represented by finite state-space closed queueing network models (continuous-time Markov processes), with finite calling population and limited servers. Various methods are used to solve the steady-state balance equations of the non-Jacksonian Markov network. Starting and stopping criteria for the solution methods are also discussed. Simulation is suggested as a viable and effective tool in solving for larger, realistic problems. Gross et al. (1987a) examine the transient behavior of Large Markovian finite source, finite repair capacity, two-echelon repairable inventory systems for a single item. They employ a randomization algorithm to reduce the Markov process transient probabilities calculation to those of a Markov chain and underlying Poisson process, both of which are amenable to exact calculation. They also show representation of models with two types of units, and models with three levels of supply and repair. Gross et al. (1987a) use a network decomposition approach for approximating the steady state behavior for the Markovian two-echelon repairable item inventory system for the case of a single item. Finite calling population and finite repair capacity are assumed, and the performance measure is availability. The approximate solution procedure for steady state probabilities decomposes the large Markovian queue-

ing network into smaller overlapping local subnetworks which are then solved. Gross and Miller (1984) model the multi-echelon repairable item provisioning system in a time-varying environment. They assume a finite population of repairable items along with limited repair capacity and focus on transient solutions to the system, obtained using a randomization technique. The approach used to obtain solutions is a two-stage procedure, the first stage is a method of modeling called SERT, and the second stage is based on the randomization technique. Exact solutions are obtained for the transient conditions. The solutions are compared with the METRIC based technique of DYNA-METRIC, which was developed for transient conditions and the advantage of this approach shown. In an earlier paper, Gross et al. (1983) developed exact models for a similar problem but for steady-state solutions only. They minimize system cost subject to specified service levels, and obtain the optimal number of servers at depot and bases along with spares levels, for the finite calling population, finite repair channels case. Again, the approach involves modeling the process using closed queueing network theory and an implicit enumeration algorithm is used to obtain the exact solution.

Gupta (1993) develops an approximate solution to the multi-indentured problem for a single base. The model considers a single item, and repair is limited to the depot. The model is formulated as a continuous-time Markov process, and assumes a finite calling population and finite servers. A FCFS rule is used for priority scheduling. The objective of the paper is to determine the steady-state characteristics of the system., and the computational results are compared with simulation, since exact solutions cannot be obtained for the model. Gupta and Albright (1992) also discuss the case for several bases and develop approximate solutions for the steady-state characteristics of the system. They assume single failures, only the spare modules are stocked and no end items, servers are assigned exclusively to modules, and a FCFS policy for return of modules to depot. They use small size test problems and compare results to those obtained via simulation.

Jung (1993) considers the case of time-varying demand for repairable inventory systems. A method for determining system characteristics under reliability improvement is presented.

10. Application-based models

Most applications models were developed and implemented in a particular setting, primarily transportation equipment such as bus fleets, locomotives, or airlines. There are a very limited number of studies which actually document the use of repairable inventory model, although METRIC was developed for the US Air Force and variants of it have been in use for some time. We cover a variety of reported applications in this section, beginning with decision support systems.

10.1. Decision support systems

Because of the complexity of the problems of spares setting, allocation of spares and control of the repair operations are very complex, several authors have developed decision support systems (DSS) to aid managers in making the best choices. Kumar et al. (1994) discuss the development of a DSS for the Indian Railroad for diesel locomotive fleet spares. The core of the DSS is based on physical systems theory and systems dynamics and was developed to guide managers in managing production, repair, storage and transportation policies. The model is limited to a single level of indenture product structure and has been implemented for a limited number of items with success. Petrovic and Petrovic (1992) report on the development and use of a expert system (SPARTA) for electronic items for both consumable spares and repairable spares. The DSS appears to be primarily designed to give heuristic recommendations for stocking decisions only. Tedone (1989) reports on a DSS developed and currently in use by American Airlines. The system is reported in very general terms, with no details on the mechanics of the models used. Tedone (1989) does however mention that the system incorporates the cost of canceled flight and that this cost was surprisingly difficult to estimate. Malmborg et al. (1987) develop a DSS designed to evaluate maintenance policies for a repairable item population, but the system was not implemented and many of the details were not based on any actual operations.

Most of the works describing DSS development were lacking in a through discussion of the underly-

ing mechanism driving the DSS, with the notable exception of Kumar et al. (1994). It is also difficult to tell exactly what benefits gained, in terms of costs and service levels, because of the limited nature of the studies and a lack of financial data. The area appears to have a great deal of under-utilized potential for managing complex repairable inventory system because of the ease of use for DSS applications and the less intimidating requirements for a potential user.

10.2. Other application models

Kharola and Vrat (1986) report on an interesting model for developed for a two-echelon bus fleet problem. The model is a heuristic designed to minimize the total systems cost and addresses the problem of initial spares setting while considering scheduled overhauls, maintenance, manpower and inventory. Kharola and Vrat (1986) report that the system was successfully implemented for a limited number of units. Singh et al. (1980) also describe a model developed for a bus fleet problem where failure rates are based on the geographic area of the bus route and the average mileage. A single product type is treated as a single indenture structure and the total system cost is minimized via marginal analysis. Howard (1984) extends standard inventory control theory to repairables. The system modeled is two-level, with repair at both levels. Failed units are batched and sent for repair to the depot in a given batch size using a modified (s,S) approach. A joint estimate of (s,Q) is performed. The paper is also a case study of the stock control of repairables for the National Coal Board in Great Britain. Kumar and Blumberg (1991) present a finite population queuing model designed to aid in minimizing the total annual cost in a mining equipment application. The model is a single echelon application and seek to determine the minimum number of heavy trucks required. Davidson (1980) reports on a comparison of models designed for initial spares provisioning for a new aircraft being used by Air Canada. While none of the models compared are discussed in great detail, it is clear that all the models use unique solution methodology, but Davidson (1980) reports that no real differences were found in the output. The choice of the model was based on the ease of use and clarity of the solution provided. Most recently, Ashayeri et al. (1996) report on the development an inventory system for the management of repairable service parts for personal computers at Olivetti. The study points out that a firm's good reputation is closely linked with the availability of spares. We strongly believe that this aspect of service, i.e., providing 100% system availability, will become an increasingly important dimension in firms' winning and keeping orders in the market.

11. Conclusion

Current trends will make the management of repairable inventory more critical in the future. Some of these trends include declining budgets (notably in the military) for repairables, environmental trends and regulations which are changing the character of products which were once considered disposables to being considered repairables through recoverable manufacturing processes. The implications of these trends include questioning of assumptions made in past models, such as large calling populations, ample repair capacity, no cannibalization, and no lateral shipments. Trends in product design such as increasing levels of modularization make the multi-indenture nature of the problem increasingly important. With aging products, condemnation rates for the repairable components also change. Repairable inventory policies must take into account condemnation rates, and develop appropriate procurement policies, along with the system for managing the repairable items which do get repaired.

There also has to be a stronger tie-in of repairable models to service levels, i.e., the availability of the items. The environmental greening effort is also leading the push towards closed-loop logistics systems. These need to be considered in the development of repairable inventory systems. Another area for development of models is decision support systems to aid managers in making the best choices. The need exists for the development of models and repairable inventory systems which incorporate real-

ism through incorporation of features and approaches as have been discussed above.

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